

Week 7 — Diagnostic Methodology

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours

Primary LOs: LO-09

Hands-on target: Session 2 ≈ 90% practical

Program spine week: Full application of the 7-step process

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [] Prepare 3–5 carts with distinct complaints (mix planted + real)
- [] Print labs/W07_seven_step_checklist.md and labs/W07_diagnostic_report.md (also Appendix-style)
- [] Print flowchart handout common_fault_flowcharts.md
- [] Assign pair teams; prepare “customer interview” cue cards for instructor role-play
- [] Ensure each cart has matching OEM diagram packet
- [] Timebox: each pair gets ~45–50 min on cart + documentation

Suggested planted scenarios (label privately)

Cart	Symptom card	Root cause
A	“Won’t move / clicks”	Weak solenoid contacts or open coil feed
B	“Slow / weak”	Marginal battery or high-R cable
C	“Dead / no power”	Main fuse/open connector / pack imbalance
D	“Charger won’t charge”	Interlock or AC supply issue
E	“Intermittent”	Loose pin / HPD sequence / throttle connector

Debrief wall

Whiteboard columns: Symptom → First tests → Decision → Actual root cause.

Week 7 Learning Objectives

1. Apply the structured 7-step diagnostic process to any golf cart complaint.
 2. Use simple troubleshooting flowcharts without skipping isolation.
 3. Document findings professionally and present recommendations clearly.
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Session 1 — Theory & Demonstration (2 hours)

Materials needed: Posted 7-step banner; handouts/common_fault_flowcharts.md; good vs bad report examples; demo cart for think-aloud; customer interview cue cards; labs/w07_diagnostic_report.md blank.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety + “no parts roulette” culture	
0:05	20	Teach 7 steps in detail with examples	Post the 7 steps permanently
0:25	15	Cost of wrong diagnosis stories (anonymized)	Battery pack replaced for a \$20 connector
0:40	15	Flowcharts: No Go / Slow / Charger / Intermittent	Walk one fully
0:55	5	Break	
1:00	20	Interview skills: history, when, weather, recent work	Role-play 2 minutes
1:20	20	Documentation standards: data, units, photos, torque	Show good vs bad report
1:40	15	Live instructor demo on one cart (compressed 7-step)	Think aloud
1:55	5	Lab pair assignments & scenario rules	No sharing answers across teams

The 7-Step Process (post in bay)

1. Verify the complaint
2. Gather information & history
3. Perform visual & safety inspection
4. Isolate systems
5. Test components
6. Perform repair
7. Verify repair & road test

Key Teaching Points

- Start with batteries and connections on electric “no go/slow” — highest probability, safest logic.
- Isolation means proving what still works, not randomly swapping.
- If you can’t explain why a part failed, you may not have the root cause.
- Verification is part of the job — not optional.

Instructor say/do (high-stakes moments)

- **Say:** “If you can’t tell me what the last measurement proved, you’re not diagnosing yet.”
- **Do:** During lab, interrupt once per team with: “What system did you just rule out?”
- **Say:** “Your report is the product — the cart running is only half.”
- **Do:** Show a one-page bad report vs good report side-by-side before pairs start.

Common Misconceptions

Misconception	Correction
“Experienced techs skip steps.”	Experts compress steps — they don’t skip evidence.
“Flowcharts are for beginners only.”	Pros use them under pressure and for training consistency.
“If the new controller fixed it, diagnosis was good.”	Intermittents and stacked faults fool parts-swappers.

Safety Reminders

- Still LOTO when required during isolation.
- Road test only after instructor clearance; designated route.
- Don’t defeat interlocks to “make it run” for a demo to customers.

Session 2 — Hands-On Lab (2 hours)

Materials needed: labs/W07_seven_step_checklist.md, labs/W07_diagnostic_report.md; 3-5 faulted carts; OEM diagram packets; full tool/PPE sets; debrief whiteboard.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety; distribute symptom cards; start timers	
0:05	10	Step 1-2: Verify + interview instructor/customer	
0:15	10	Step 3: Visual & safety	
0:25	35	Steps 4-5: Isolate & test (pairs)	Instructor circulates; ask "what did that reading prove?"
1:00	5	Buffer	
1:05	20	Step 6-7: Repair (if authorized) + verify	Some teams stop at recommend if parts unavailable
1:25	25	Group debrief presentations (3-4 min/pair)	Score process quality, not only luck
1:50	10	Collect reports; cleanup; preview mechanical week	

Assessment Focus

Diagnostic process checklist completion + quality of presentation/documentation (use documentation rubric in Program Guide).

Week 7 Quiz (8 questions)

- Step 1 of the 7-step process is:
 - Order parts
 - Verify the complaint
 - Replace the controller
 - Skip to road test
- The best reason to gather history is:
 - Gossip
 - Recent work, conditions, and intermittency patterns guide testing
 - To avoid measuring anything
 - To justify skipping LOTO
- System isolation means:
 - Testing randomly
 - Narrowing which subsystem can produce the symptom using logic and measurements
 - Removing all fuses permanently
 - Painting the frame

4. Professional documentation should include:
- A) Only “fixed”
 - B) Tests, readings with units, diagnosis, actions, verification
 - C) Customer home address and SSN
 - D) Memes
5. True or False: Verification/road test can be skipped if the tech is confident.

Short Answer

1. Write the 7 steps in order.
2. Why start many electric drivability diagnoses at the batteries/connections?
3. Give one example of confirming a repair after replacing a solenoid.

Answer Key

1. **B**
2. **B**
3. **B**
4. **B**
5. **False**
6. Verify → Gather info → Visual/safety → Isolate → Test → Repair → Verify/road test
7. High failure rate / foundation of voltage delivery; prevents mis-condemning controllers/ motors
8. Confirm pack voltage at controller under load, no excessive contact drop, cart accelerates normally, no codes, road test OK